

WAGA Flarm Tool & Simulator User Notes

FOR VERSION 310326

VERSION RECORD

Ver 271025	Initial version.
Ver 310326	Much improved Flarm configuration tool, including CARP data read.

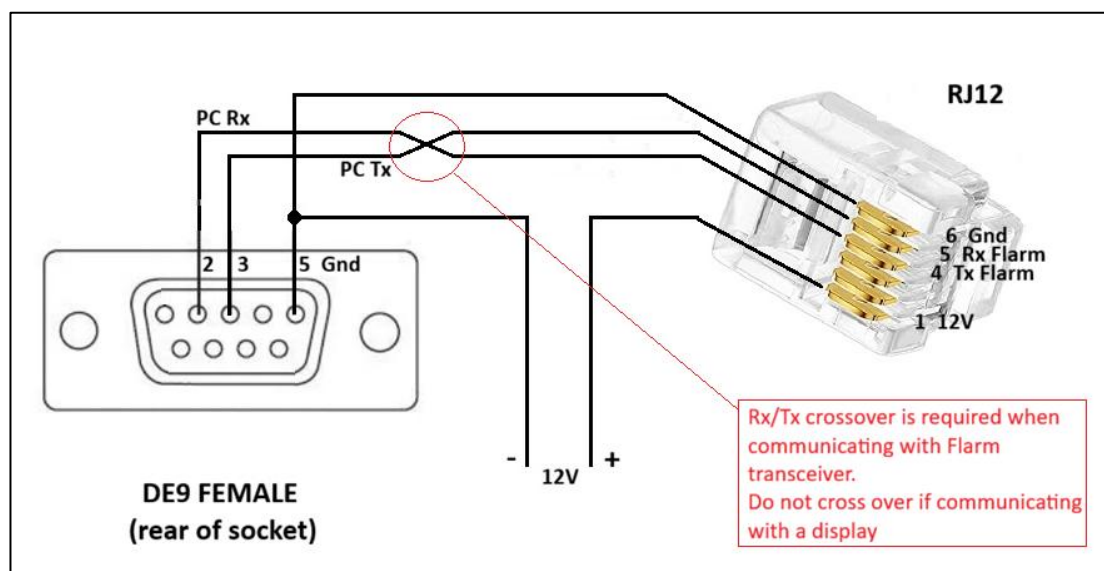
INTRODUCTION

The Flarm Tool and Simulator (the Tool) is intended to be a tool to assist developers of Flarm or SoftRF displays. It was built for testing the SkyView WAGA Display but is likely to be useful for other displays. It's also a convenient tool for reading and changing Flarm configuration settings and extracting CARP receive performance data.

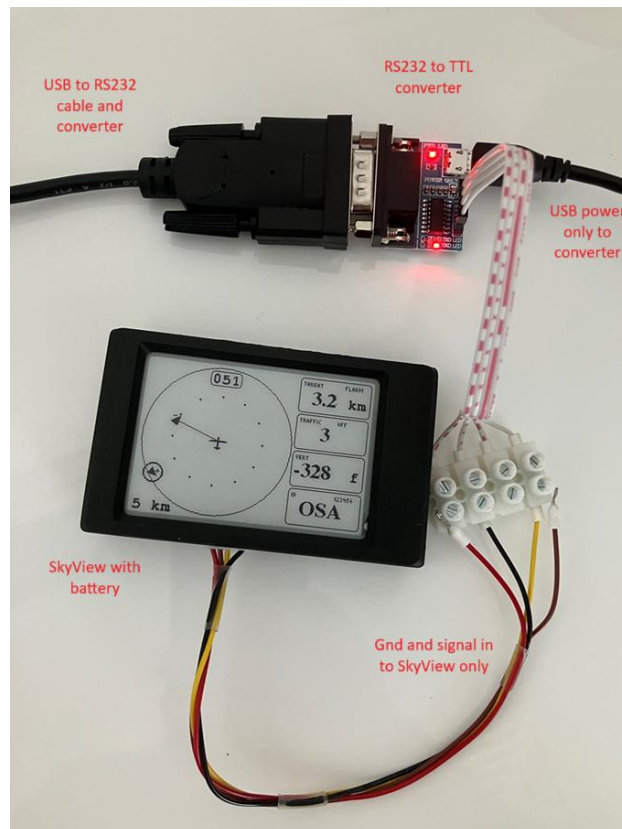
The Tool is built using MS Access and VBA coding. This platform was chosen due to the database intensive nature of the simulations and the ease of building a user interface. It is a rough tool, not a 'polished' programme. Some understanding of MS Access VBA is needed to set up custom simulations and change parameters.

Flarm RJ12/45 connections require data in RS232 protocol. Connect to a PC using a USB-RS232 convertor-DE9 cable. From the DE9 connector:

- To connect to a Flarm transceiver, use a standard IGC Flarm DE9 to RJ12/RJ45 cable (as shown in Classic Flarm manuals) which has Rx/Tx crossover.
- To use with a Flarm display, the standard cable must not have Tx/Rx crossed over.



- To connect to a SkyView display, use a RS232 to TTL converter. See photo below.



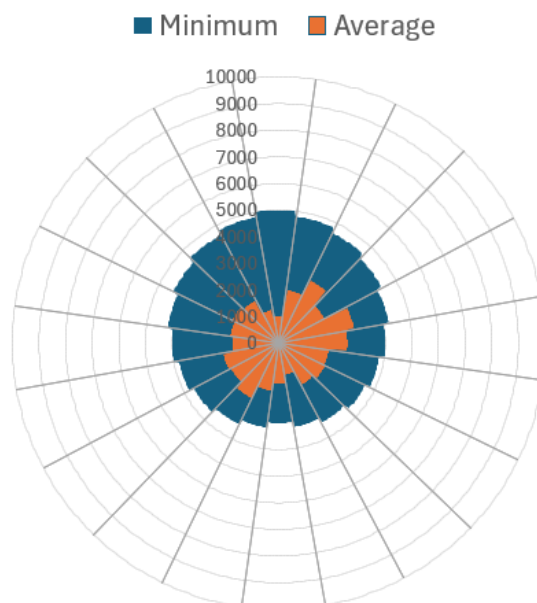
FLARM TOOL

The Flarm Tool screen can be used to read and set the configuration settings of Classic, PowerFlarm and Atom Flarm devices (including the WAGA Flarm). The settings are described in Flarm document FTD-014.

The configuration read sequence includes reading the Continuous Analyzer of Radio Performance (CARP) data but currently only antenna A. There is no data reset button provided, but this can easily be added or achieved with a serial terminal programme (eg Putty) by sending the \$PFLAN,S,RESET command. CARP Rx analysis is described in FTD-065.

There is an Excel workbook in this GitHub repository that draws the Flarm style 'filled' radar graph. Just copy the 'CARP range' data from the Access tool into the Excel workbook indicated cell.

Note; Formula may need some adjustments if the range data is more than 4 digits.



WAGA FLARM TOOL - CONFIG & CARP

Read Settings

This PC port settings for Flarm

Select Port Auto Find Flarm WAGA Flarm default 57600

Select Baud Set

Port Bit Set 8-N-1 Should be 8-N-1

K Wilson 31/03/26

Auto Find Flarm Port finds the first port in use (ports 1-10 only). If in doubt, use Device Manager to look up USB port nbr

System Info (read only)

SER

SWVER

RADIOID As transmitted

DEVTYPE

Classic Flarm Settings

Baud Set

NMEA OUT Set Protocol / Data content

PowerFlarm CARP Data

Start

End

Stats Count

CARP range antenna A

RFTOP,A,3627,4306,4131,3990,3917,4014,3811,3634,3393,2877,2730,3370,3587,3406,3319,3247,3147,3138,3351,3409

CARP = Continuous Analyser of Radio Performance.
See FLARM FTD-065
To reset: \$PFLAN,S,RESET

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PowerFlarm Built-In Simulations Driver

Simulations From Data Tables

Note: Set ID to ICAO hex, or to FFFFFFFF to use Flarm device ID

The correct device port and baud rate must be selected for successful communication.

POWERFLARM BUILT-IN SIMULATIONS DRIVER

PowerFlarms have 6x built-in traffic simulations, each intended to last approx 30 seconds. If a PowerFlarm is connected to the SkyView Display, then the Tool can issue the required \$PFLAF commands to the PowerFlarm that will cause it to commence outputting the selected simulation.

In this mode, you are not using the simulations contained in the data tables, you are initiating the PowerFlarm's built-in simulations.

WAGA FLARM TOOL - POWERFLARM BUILT-IN SIM DRIVER

K Wilson 2025

Flarm's Built-In Simulations

Sim 1

Sim 2

Sim 3

Sim 4

Sim 5

Sim 6

1 A single **FLARM-equipped aircraft** with ID 123456 in a collision trajectory with 0° relative bearing. Starts far away with no warning and goes through all alarm levels until collision. Lasts 30 seconds.

2 A single **ADS-B-equipped aircraft** with ID 123456 in a collision trajectory with 270° relative bearing. Starts far away with no warning and goes through all alarm levels until collision. Lasts 30 seconds.

3 A single **non-directional aircraft** (equipped with a Mode-S transponder) with ID 123456 in a collision trajectory. Starts far away with no warning and goes through all alarm levels until collision. Lasts 30 seconds.

4 A **fixed obstacle** from an installed obstacle database with a valid license with ID 123456. Starts far away with no warning and goes through all alarm levels until collision. Lasts 30 seconds.

5 An **Alert Zone**, i.e. a dynamic airspace in the form of a cylinder where special vigilance is required (e.g. active skydiving activity) with ID 123456. Ownship flies towards the zone for 4 seconds before entering, crosses for 24 seconds, and continues on the other side for 2 seconds. Lasts 30 seconds.

6 A **mixed scenario**, where multiple traffic types and objects are combined. A FLARM-equipped aircraft (ID 123456), an ADS-B-equipped aircraft (ID 123457), a non-directional aircraft (ID 123458), a fixed obstacle (ID 123459), and an alert zone (ID 123460) are all in the vicinity, none of which are generating a warning. Lasts 30 seconds.

The buttons on this menu will send the appropriate \$PF... command to a connected PowerFlarm, causing the PowerFlarm (not this simulator) to send the out the Built-In traffic simulations.

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Configuration & CARP

Simulations From Data Tables

Set port and baud rate on Main Form

FLARM SIMULATOR

Simulations are constructed as Flarm data sentences (\$PFLAU and \$PFLAA) stored as records in Access data tables. The Sim-Tx Form in the Access database creates a time variable 'pulse' that triggers the sending of data sentences sequential read from the selected simulation data table. Simulations loop continuously. The data is output to a selected PC port.

GPS fix

The SkyView requires GPS fix data (\$GPGGA and \$GPRMC) in order to activate the radar screen. GPS data can be provided by:

Embedded as discrete records in the simulation tables, or;

The Tool can interleave GPS data for a pre-set static location with the Flarm sentences as they are sent. . Both GPS sentence types are sent in one trigger cycle.

ID	Data
1	\$GPGGA,120000,4518.8701,N,07550.2500,W,1,12,0.92,1000.0,M,-34.3,M,,*75
2	\$GPRMC,120000,A,4518.8701,N,07550.2500,W,0.0,0.0,100223,,,A*65
3	\$PFLAU,1,1,2,1,0,333,0,-50,2236,C043EC*21
4	\$PFLAA,0,2000,-1000,-50,1,C043EC,0.00,0.00,0.00,0.00,1*32
5	\$GPGGA,120001,4518.8779,N,07550.2363,W,1,12,0.92,1001.0,M,-34.3,M,,*79
6	\$GPRMC,120001,A,4518.8779,N,07550.2363,W,23.1,51.0,100223,,,A*6C
7	\$PFLAU,1,1,2,1,0,282,0,-51,2251,C043EC*2A
8	\$PFLAA,0,2010,-1011,-51,1,C043EC,334.00,-26.00,2256.00,0.50,1*19
9	\$GPGGA,120002,4518.8833,N,07550.2202,W,1,12,0.92,1002.0,M,-34.3,M,,*7E
10	\$GPRMC,120002,A,4518.8833,N,07550.2202,W,23.1,63.0,100223,,,A*69
11	\$PFLAU,1,1,2,1,0,270,0,-52,2265,C043EC*23
12	\$PFLAA,0,2022,-1020,-52,1,C043EC,28.00,54.00,26.00,0.50,1*08
13	\$GPGGA,120003,4518.8867,N,07550.2031,W,1,12,0.92,1003.0,M,-34.3,M,,*7D
14	\$GPRMC,120003,A,4518.8867,N,07550.2031,W,23.1,75.0,100223,,,A*6C
15	\$PFLAU,1,1,2,1,0,258,0,-53,2279,C043EC*25
16	\$PFLAA,0,2035,-1025,-53,1,C043EC,42.00,14.00,26.00,0.50,1*02
17	\$GPGGA,120004,4518.8872,N,07550.1855,W,1,12,0.92,1004.0,M,-34.3,M,,*70
18	\$GPRMC,120004,A,4518.8872,N,07550.1855,W,23.1,87.0,100223,,,A*6B
19	\$PFLAU,1,1,2,1,0,246,0,-54,2292,C043EC*28

*Example Simulation table with GPS data embedded and each sentence ends with a CheckSum (the 2 characters after the *)*

Limitations

The simulations created with the Tool must be regarded as approximations only. It can be used for demonstrations, but the resulting radar displays may distort time, distance and apparent traffic path and speed. The major limitations are:

- ThisAircraft is effectively stationary, unless real recorded or meticulously created data is used in the tables. This means the apparent traffic paths are not truly realistic other than when headed directly towards or away from ThisAircraft.

- The Tool sends data sentences at regular intervals whereas, in reality, data is received at the SkyView most likely in an irregular stream eg as \$PFLAA data is received from multiple different traffic targets. There may be 3 say \$PFLAA sentences for every \$PFLAU.

Checksum

The SkyView requires that each data sentence has a valid CheckSum else it will ignore the data. An example CheckSum looks like *68 where the number 68 is the actual CheckSum value. The Tool has a CheckSum calculator which can append the required value if it is missing from the data sentence.

There are numerous online CheckSum calculators eg <https://nmeaChecksum.eqth.net/>

Flarm data sentences

There are two main sentence types used to pass traffic data and alarms which start with the keys \$PFLAU and \$PFLAA.

\$PFLAU. This is a once a second heartbeat from the transceiver and the primary conveyor of threat/alarm information. Traffic is located by relative bearing and distance from ThisAircraft. Plus relative vertical height above/below.

\$PFLAA. This passes data on other proximate aircraft, including alarm information. Traffic position is given by relative distance 'north' and 'east' but relative to ThisAircraft track, so really it is in front/behind and left/right. Plus relative vertical height above/below. The 'Source' field can determine if the traffic is Flarm, ADSB or Mode-S.

Definitions of the sentence structures and parameters can be found in Flarm documents FTD-012 and FTD-014. All these parameters can be set or modified in the custom simulations. Take care to understand the fields formats and units.

PowerFlarm Built-in simulations using data tables

The 6x PowerFlarm built-in simulations have been recorded and are included as Tables in the simulation database. It is thus not necessary to connect the SkyView to a Flarm in order to use the PowerFlarm sims.

Since the records in these tables include a CheckSum, then following any modification to the fields in these records, the CheckSum must be calculated and updated. It would be easier, to strip the CheckSum from the data and then use the tool's CheckSum function to automatically calculate and append the CheckSum as the data is sent.

Custom simulations

The tool can save a large number of custom created simulations. The simulations can be constructed from a single type of Flarm sentence or combining the 2 types.

Examples are included in the database where Single Flarm sentence type simulations can be constructed directly in the MS Access tables by entering data into cells for each field and

the programme automatically concatenates all the rows fields into a CSV sentence in the 'Data' field/column (without CheckSum).

Mixed Flarm Sentence types are best created by the following steps:

1. Create an MS Access table for each sentence type
2. Copy/paste the calculated/concatenated field of each table to separate excel sheets. Do not separate out the fields.
3. Add a column alongside the data and insert a numbering scheme that will be used to sort the data when the two files are merged. Eg if alternating \$PFLAU with \$PFLAA records, then give odd numbers to the PFLAU records and even numbers to the \$PFLAA records.
4. Append the \$PFLAA records (with the numbering) to the end of the \$PFLAU data.
5. Sort by the number column.
6. Create a new data table in the MS Access tool and copy the merged data into this table.

Simulations can be constructed as either:

Sequential lines. Sentences are entered in every record of the sim table and sent at the rate of 1/SimDelay. Ideally the \$PFLAU sentence should be sent once per second. If the Tool is set (FixNeeded=True) to add GPS data, this will interleave the Flarm sentences every other trigger so make allowance for this. Eg set SimDelay=250 (4 triggers in 1 second);

ID	Time	Sentence
1	0	\$PFLAU, ...
2	250	\$GPGGA, ... and \$GPRMC, ...
3	500	\$PFLAA, ...
4	750	\$GPGGA, ... and \$GPRMC, ...
5	1000	\$PFLAU, ...
6	1200	\$GPGGA, ... and \$GPRMC, ...
cont..		

Time frames. This method establishes a number of time slots per second. Eg set SimDelay=100ms so that there are slots for 10 sentences per second. In a sim table enter \$PFLAU sentences every 10th record (eg 1 per second). Enter GPS data strings in slots 2 and 3. Enter \$PFLAA strings in any of the free slots and leave the remainder empty.

Simulation settings

Each simulation must be set up I the SimSetting Table. The SimLetter can be any number, letter or combination. The SimTable field entry must exactly match a simulation Table in the database. More simulations can be added as desired.

ID	SimLetter	SimTable	SimDelay	SimFix	SimCSur	Description
2	A	Sim_PF6_noObstacle	250	☒	☐	PowerFlarm Sim 6 without zone and obstacle.
3	B	Sim_Moshe	250	☐	☐	Moshe -3 aircraft.
4	C	PFLAU	250	☒	☒	Basic alarms. Needs fix and Csum
5	D	PFLAA	250	☒	☒	Needs fix and Csum
6	E	2Therm_B	250	☒	☒	2 gliders thermalling around
7	F	Moshe_short	1000	☐	☐	
8	G	PFLAA-ModeS	1000	☒	☒	Mode S
9	H		0	☒	☐	TBC
10	I	Moshe_Original	500	☐	☐	Moshe 2 sim original
11	J	Tug_Sim_PFLAU	1500	☒	☒	Tug with glider on tow, then threat
12	K		0	☐	☐	TBC
※ (New)			0	☐	☐	

The SimDelay (interval) can be modified in the GUI panel and changes saved in this table.

Built-in GPS data

GPS data sentences (\$GPGGA and \$GPRMC) for a static position are hard-coded in VBA 'module 1'. This data can be edited provided the correct format is maintained. The code automatically adds the correct CheckSum.

```
Public Sub SendFix()
Dim i As Integer
' ThisAircraft fix data is required if not in sim Table file.
' Any changes to the GPS lines requires recalculation of the CheckSum (the *xx at end of the string),
' which is done automatically. Can be checked using calculator at: https://nmeachecksum.eqth.net/
' ThisAircraft_Track variable in $GPRMC sentence sets the course over the ground. Entered in the GUI.

i = 0
FixData = "$GPGGA,120001,3200.0000,S,11725.0000,E,1,12,0.92,1001.0,M,-34.3,M,,*"
FixData = FixData & NMEAChecksum(FixData) & Chr(13) & Chr(10)

For i = 1 To Len(FixData)
Call PUT_COM_PORT(SkyViewPort, Mid(FixData, i, 1))
Next i

' Example "$GPRMC,120001,A,3200.0000,S,11725.0000,E,23.1,51.0,100223,,A*68" & Chr(13) & Chr(10)
i = 0
ThisAircraft_Track = Forms!Simulations!Txt_Track

FixData = "$GPRMC,120001,A,3200.0000,S,11725.0000,E,23.1," & ThisAircraft_Track & ",100223,,A*"
FixData = FixData & NMEAChecksum(FixData) & Chr(13) & Chr(10)
Debug.Print FixData

For i = 1 To Len(FixData)
Call PUT_COM_PORT(SkyViewPort, Mid(FixData, i, 1))
Next i

End Sub
```

GPS aircraft track

In the Custom simulations panel, you can set This>Aircraft track angle (0-360 degrees). If testing with the radar display in track-up mode, the traffic position(s) on the radar will not change as Flarm sentences provided position data relative to ThisAircraft heading.

Annex A. Extract from Flarm Document FTD-012

8.1 PFLAU – Heartbeat, status, and basic alarms

Syntax:

PFLAU,<RX>,<TX>,<GPS>,<Power>,<AlarmLevel>,<RelativeBearing>,<AlarmType>,<RelativeVertical>,<RelativeDistance>[,<ID>]

Description:

Heartbeat message; output once per second. Consumers should use this message to detect the presence (and absence) of a compatible data stream.

The sentence summarizes the most relevant status information from the last one-second interval: RF status (RX, TX), power state, and the most important current threat, either traffic, an obstacle, or an alert zone. Consumers with limited resources (e.g. with respect to display capabilities or computational resources) can thus use PFLAU to display basic safety information. Other consumers shall also use PFLAA for extended information.

On devices with SSR/ADS-B Module, non-directional targets are output if enabled (PCASPFLAU configuration setting).

For data port version ≥ 7 , Alert Zone alarms are available; see the <AlarmLevel> and <AlarmType> fields.

For data port version ≥ 4 , traffic advisory notifications (INFO alarms) are available, see the <AlarmType> field.

8.2 PFLAA – Data on other proximate aircraft

Syntax:

PFLAA,<AlarmLevel>,<RelativeNorth>,<RelativeEast>,<RelativeVertical>,<IDType>,<ID>,<Track>,<TurnRate>,<GroundSpeed>,<ClimbRate>,<AcftType>[,<NoTrack>[,<Source>,<RSSI>]]

Description:

Data on other proximate aircraft, intended for connected devices with sufficient CPU performance. This sentence should be treated with utmost flexibility and tolerance on a best effort base. Individual parameters may be empty. The sentence is only sent when port baud rate is 19.2k or higher. In case of serial port congestion or high CPU load, this sentence may be omitted for several objects independent of the alarm level. On devices with SSR/ADS-B Module, ADS-B and non-directional targets are output as well (transponder Mode-C/S only from protocol version 6 and higher).

Obstacle information is not delivered with this sentence.

Note that in case of many targets within range, individual targets, including the most dangerous one, might not be delivered every second, not regularly, or not at all, due to less strict priority handling for the PFLAA sentence. **Always use PFLAU as primary alarm source.** Usually, but not always, the last PFLAA sentence is the one causing the PFLAU content. The other PFLAA sentences are not ordered. Do not expect to receive PFLAU <Rx> times PFLAA sentences, because the number of aircraft being processed might be higher or lower. PFLAA sentences can be based on extrapolated historical data. PFLAA sentences are limited to other aircraft with a horizontal and vertical distance less than the configured range. On Classic FLARM-based devices, the vertical distance is always 500 m. Non-moving aircraft are suppressed.

Annex B. \$PFLAA and \$PFLAU example simulation tables

Note that the data column is the concatenation of the other columns with commas inserted between fields and no CheckSum. Only the 'Data' column is sent as a sentence. The CheckSum is added by the Tool provided CheckSum=True is set.

ID	PFLAA	AlarmLev	Rel_N	Rel_E	Rel_V	IDT	TargetID	Track	GS	ClimbRate	ActTyp	NoTr	Source	Data
1	\$PFLAA	0	0	3500	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,0,0,3500,373,1,DD506F,270,,150,0,2,0,0,
2	\$PFLAA	0	0	3250	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,0,0,3250,373,1,DD506F,270,,150,0,2,0,0,
3	\$PFLAA	0	0	3000	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,0,0,3000,373,1,DD506F,270,,150,0,2,0,0,
4	\$PFLAA	0	0	2750	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,0,0,2750,373,1,DD506F,270,,150,0,2,0,0,
5	\$PFLAA	1	0	2500	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,1,0,2500,373,1,DD506F,270,,150,0,2,0,0,
6	\$PFLAA	1	0	2250	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,1,0,2250,373,1,DD506F,270,,150,0,2,0,0,
7	\$PFLAA	1	0	2000	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,1,0,2000,373,1,DD506F,270,,150,0,2,0,0,
8	\$PFLAA	2	0	1750	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,2,0,1750,373,1,DD506F,270,,150,0,2,0,0,
9	\$PFLAA	2	0	1500	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,2,0,1500,373,1,DD506F,270,,150,0,2,0,0,
10	\$PFLAA	2	0	1250	373	1	DD506F	270	150	0	2	0	0	\$PFLAA,2,0,1250,373,1,DD506F,270,,150,0,2,0,0,
11	\$PFLAA	3	0	1000	0	1	DD506F	270	150	0	2	0	0	\$PFLAA,3,0,1000,0,1,DD506F,270,,150,0,2,0,0,
12	\$PFLAA	3	0	750	-373	1	DD506F	270	150	0	2	0	0	\$PFLAA,3,0,750,-373,1,DD506F,270,,150,0,2,0,0,
13	\$PFLAA	3	0	500	-373	1	DD506F	270	150	0	2	0	0	\$PFLAA,3,0,500,-373,1,DD506F,270,,150,0,2,0,0,
14	\$PFLAA	3	0	250	-373	1	DD506F	270	150	0	2	0	0	\$PFLAA,3,0,250,-373,1,DD506F,270,,150,0,2,0,0,
15	\$PFLAA	1	0	0	-373	1	DD506F	270	150	0	2	0	0	\$PFLAA,1,0,0,-373,1,DD506F,270,,150,0,2,0,0,
16	\$PFLAA	1	0	-250	-373	1	DD506F	235	150	0	2	0	0	\$PFLAA,1,0,-250,-373,1,DD506F,235,,150,0,2,0,0,
17	\$PFLAA	1	-250	-500	-373	1	DD506F	235	150	0	2	0	0	\$PFLAA,1,-250,-500,-373,1,DD506F,235,,150,0,2,0,0,
18	\$PFLAA	3	-500	-750	-373	1	DD506F	235	150	0	2	0	0	\$PFLAA,3,-500,-750,-373,1,DD506F,235,,150,0,2,0,0,
19	\$PFLAA	3	-750	-1000	-373	1	DD506F	235	150	0	2	0	0	\$PFLAA,3,-750,-1000,-373,1,DD506F,235,,150,0,2,0,0,
20	\$PFLAA	3	-1000	-1250	-373	1	DD506F	235	150	0	2	0	0	\$PFLAA,3,-1000,-1250,-373,1,DD506F,235,,150,0,2,0,0,
21	\$PFLAA	0	-1250	-1500	-373	1	DD506F	235	150	0	2	0	0	\$PFLAA,0,-1250,-1500,-373,1,DD506F,235,,150,0,2,0,0,

ID	PFLAU	RX	TX	gPS	Pow	AlarmLevel	Rel_B	AlarmTyp	Rel_V	Rel_D	TargetID	Data
1	\$PFLAU	2	1	0	0	0	90		0	4500	DF0957	\$PFLAU,2,1,0,0,0,90,,0,4500,DF0957
2	\$PFLAU	2	1	0	0	0	90		0	4000	DF0957	\$PFLAU,2,1,0,0,0,90,,0,4000,DF0957
3	\$PFLAU	2	1	0	0	0	90		0	3500	DF0957	\$PFLAU,2,1,0,0,0,90,,0,3500,DF0957
4	\$PFLAU	2	1	0	0	0	90		0	3000	DF0957	\$PFLAU,2,1,0,0,0,90,,0,3000,DF0957
5	\$PFLAU	2	1	0	0	0	90		0	2500	DF0957	\$PFLAU,2,1,0,0,0,90,,0,2500,DF0957
6	\$PFLAU	2	1	0	0	0	90		0	2000	DF0957	\$PFLAU,2,1,0,0,0,90,,0,2000,DF0957
7	\$PFLAU	2	1	0	0	0	90		0	1750	DF0957	\$PFLAU,2,1,0,0,0,90,,0,1750,DF0957
8	\$PFLAU	2	1	0	0	1	90		0	1500	DF0957	\$PFLAU,2,1,0,0,1,90,,0,1500,DF0957
9	\$PFLAU	2	1	0	0	1	90		0	1250	DF0957	\$PFLAU,2,1,0,0,1,90,,0,1250,DF0957
10	\$PFLAU	2	1	0	0	2	90		0	1000	DF0957	\$PFLAU,2,1,0,0,2,90,,0,1000,DF0957
11	\$PFLAU	2	1	0	0	2	90		0	750	DF0957	\$PFLAU,2,1,0,0,2,90,,0,750,DF0957
12	\$PFLAU	2	1	0	0	3	90		0	500	DF0957	\$PFLAU,2,1,0,0,3,90,,0,500,DF0957
13	\$PFLAU	2	1	0	0	3	90		0	250	DF0957	\$PFLAU,2,1,0,0,3,90,,0,250,DF0957
14	\$PFLAU	2	1	0	0	3	90		0	0	DF0957	\$PFLAU,2,1,0,0,3,90,,0,0,DF0957
15	\$PFLAU	2	1	0	0	0	270		0	250	DF0957	\$PFLAU,2,1,0,0,0,270,,0,250,DF0957
16	\$PFLAU	2	1	0	0	0	270		-100	500	DF0957	\$PFLAU,2,1,0,0,0,270,,-100,500,DF0957
17	\$PFLAU	2	1	0	0	0	270		-100	1000	DF0957	\$PFLAU,2,1,0,0,0,270,,-100,1000,DF0957
18	\$PFLAU	2	1	0	0	0	270		-85	1500	DF0957	\$PFLAU,2,1,0,0,0,270,,-85,1500,DF0957
19	\$PFLAU	2	1	0	0	0	270		-65	2000	DF0957	\$PFLAU,2,1,0,0,0,270,,-65,2000,DF0957
20	\$PFLAU	2	1	0	0	0	270		-45	4500	DF0957	\$PFLAU,2,1,0,0,0,270,,-45,4500,DF0957
21	\$PFLAU	2	1	0	0	0	270		-30	5000	DF0957	\$PFLAU,2,1,0,0,0,270,,-30,5000,DF0957
22	\$PFLAU	2	1	0	0	0	270		0	5500	DF0957	\$PFLAU,2,1,0,0,0,270,,0,5500,DF0957
23	\$PFLAU	2	1	0	0	0	270		45	6000	DF0957	\$PFLAU,2,1,0,0,0,270,,45,6000,DF0957
24	\$PFLAU	2	1	0	0	0	270		80	6500	DF0957	\$PFLAU,2,1,0,0,0,270,,80,6500,DF0957
25	\$PFLAU	2	1	0	0	0	270		120	7000	DF0957	\$PFLAU,2,1,0,0,0,270,,120,7000,DF0957
26	\$PFLAU	2	1	0	0	0	270		160	7500	DF0957	\$PFLAU,2,1,0,0,0,270,,160,7500,DF0957

Annex C. Simulations Form – the GUI

WAGA FLARM TOOL - SIMULATIONS FROM DATA TABLES

K Wilson 2025

PowerFlarm Recorded Simulations

Sim 1

Sim 2

Sim 3

Sim 4

Sim 5

Sim 6

1	A single FLARM-equipped aircraft with ID 123456 in a collision trajectory with 0° relative bearing. Starts far away with no warning and goes through all alarm levels until collision. Lasts 30 seconds.
2	A single ADS-B-equipped aircraft with ID 123456 in a collision trajectory with 270° relative bearing. Starts far away with no warning and goes through all alarm levels until collision. Lasts 30 seconds.
3	A single non-directional aircraft (equipped with a Mode-S transponder) with ID 123456 in a collision trajectory. Starts far away with no warning and goes through all alarm levels until collision. Lasts 30 seconds.
4	A fixed obstacle from an installed obstacle database with a valid license with ID 123456. Starts far away with no warning and goes through all alarm levels until collision. Lasts 30 seconds.
5	An Alert Zone , i.e. a dynamic airspace in the form of a cylinder where special vigilance is required (e.g. active skydiving activity) with ID 123456. Ownship flies towards the zone for 4 seconds before entering, crosses for 24 seconds, and continues on the other side for 2 seconds. Lasts 30 seconds.
6	A mixed scenario , where multiple traffic types and objects are combined. A FLARM-equipped aircraft (ID 123456), an ADS-B-equipped aircraft (ID 123457), a non-directional aircraft (ID 123458), a fixed obstacle (ID 123459), and an alert zone (ID 123460) are all in the vicinity, none of which are generating a warning. Lasts 30 seconds.

Custom Created Simulations

Sim

Sim	Sim Table used	Interval	Fix	Csum	Description
A	Sim_PF6_noObstacle	250	☒	☐	PowerFlarm Sim 6 without zone and obstacle.
B	Sim_Moshe_no_Csum	250	☐	☒	Moshe -3 aircraft.
C	PFLAU	1100	☒	☒	Basic alarms. Needs fix and Csum
D	PFLAA	300	☒	☒	Needs fix and Csum
E	2Therm_B	400	☒	☒	2 gliders thermalling around
F	Moshe_short	1000	☐	☐	
G	PFLAA-ModeS	1000	☒	☒	Mode S

Click on the blue 'Sim' button to run the sim. Use the sub-form scroll bar to view more sims.

The trigger interval (ms) can be changed in this panel.

Custom simulations are setup in the SimSettings Table

When a fix is added, set ThisAircraft track (0-360 degrees)

51

This PC port setting for SkyView

Select Port 6 Auto Find Port

Select Baud 19200 v

Try Auto Find Port if only one ported device present. Else use Device Manager to find SkyView's PC port number. Port numbers vary between SkyView devices. Baud rate must match SkyView serial port, Default=19200

✖

Configuration and CARP

PowerFlarm Built-In Simulations Driver